

ANTHONY ZHAI

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EDUCATION

Princeton University

B.A. in Mathematics + Computer Science

Princeton, NJ

September 2023 - May 2027

Relevant Coursework: Data Structures and Algorithms, Intro to Programming Systems, Intro to Theoretical Machine Learning, Reinforcement Learning, Probability and Stochastic Systems

GPA: 3.8/4.0

EXPERIENCE

Databricks

Software Engineer Intern

Mountain View, CA

Jun 2026 - Present

- Designed, implemented, and rolled out a Kubernetes CronJob to replace a legacy workspace garbage collection service, reducing p95 workspace deletion latency by $65\times$, accelerating handling of already-deleted workspace data by $1000\times$, and clearing 20M+ rows of stale + backlogged records across 100k+ cancelled production workspaces per month.
- Improved reliability and operational control through workload isolation, on-demand purging, granular releases and rollback, and a full observability suite with metrics, dashboards, logs, and on-call alerting.

Predigle

Software Engineer Intern

May - Aug 2023, 2024, 2025

- Architected and developed a full-stack geospatial web application (Angular, Leaflet, Django) for visualizing multi-layer data with dynamic toggling, shape-drawing, and real-time layer generation — initially applied to optimize coconut supply chain logistics, and now being productized for broader use cases such as visualizing localized market performance, competitor locations, and customer proximity.
- Co-developed a LangChain-based Retrieval-Augmented Generation (RAG) chatbot using open-source LLMs to help automate prior authorization assessment, enabling healthcare teams to query and summarize complex medical PDFs with higher accuracy and efficiency for prescription fulfillment.
- Built ML and LLM/OCR pipelines for insurance claim overpayment classification, achieving 97% accuracy on 400k+ claim records, identifying \$250k+ in new claims, and unlocking 150k+ previously unlabeled scanned-document data points.

Princeton University Power Electronics Research Lab

Machine Learning Research Intern

Jun - Aug 2022

- Developed a novel neural network to model magnetic core loss that uses the Fast Fourier Transform algorithm to more efficiently encode time-series data, using $<20\%$ the parameters of LSTM networks with $<15\%$ training time and equal performance.
- Analyzed downsampling methods for frequency and peak flux density to assess the relative significance of variables in predicting core loss and to reduce the parameter search space by 20%.

PROJECTS

CGBNet: A Deep Learning Framework for Compost Classification *Python, Tensorflow, Keras, Computer Vision, Deep Learning, Transfer Learning*

Co-first authored research article for **CGBNet**—a deep learning framework using computer vision to classify nitrogen-rich and carbon-rich compost for automating composting—published in IEEE Access, an engineering application journal (3.9 impact factor and 30% acceptance rate).

Visionary *Python, Django, Django-Rest-Framework, Flutter, Dart, Tensorflow, Keras, NLP*

Developed the **Visionary** mobile app to help the visually impaired interact with the world—generates narrated descriptions of the user's surroundings using text OCR, object detection, and NLP algorithms.

SKILLS

Programming Languages: Python, Java, C++, Javascript, Dart, HTML / CSS, SQL

Libraries: Tensorflow, PyTorch, Numpy, Pandas, Flutter, React, Node, Vue, Flask, Django, OpenCV

Technologies: Git, Bash, AWS, GCP, Docker, PostgreSQL, Firebase, Figma, NPM, UNIX

AWARDS

Most Innovative Startup Idea

HackPrinceton

Created EduLecture.ai, a platform for generating personalized lecture videos using only Google Slide presentations. Built with a React.js frontend and a Rest API backend powered by voice-generative AI tools, EduLecture.ai bridges the accessibility gap for remote education.